RISHI THAKUR

PYTHON DEVELOPER

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PROFESSIONAL SUMMARY

Python Developer with strong computer science fundamentals (M.Tech CSE, GATE CS qualified) and demonstrated experience developing automated data pipelines, backend web services, and relational database architectures. Proficient in object-oriented programming (OOP), asynchronous data processing, and integrating MariaDB, MySQL, PostgreSQL, and REST APIs. Skilled in writing performant code, test-driven development (TDD), and database schema reconciliation.

SKILLS

- **Core Python & Development:** Python (Advanced OOP, Multiprocessing, Concurrency, Itertools), RESTful APIs (FastAPI, Flask), PyTest, Unit Testing, Modular Design, Code Refactoring
- **Data Processing & Libraries:** Pandas, NumPy, SQLAlchemy, Scikit-learn, Requests, JSON/CSV Parsing, Data Cleansing Pipelines, Regular Expressions (Regex), Structured Logging
- **Databases & Systems:** Advanced SQL (MySQL, MariaDB, OracleDB, T-SQL) — Joins, CTEs, Indexing, Schema Modeling, Linux/Bash Scripting, Git/GitHub, Docker Basics
- **Communication & Spoken Skills:** Technical Documentation, Code Review Communication, Agile Sprint Demos, Cross-Functional Engineering Collaboration, API Contract Alignment

WORK EXPERIENCE

Chhattisgarh Infotech Promotion Society (CHiPS) | Raipur, India | 06/2025 – 03/2026
Python & Data Engineering Intern • Dept. of Health & Family Welfare, Govt. Of Chhattisgarh

- Engineered automated **Python ETL scripts** integrating biometric punch logs (AEBAS) with State Treasury payroll databases across **33 districts**, handling complex data joins.
- Developed an automated data reconciliation pipeline bridging attendance IDs to Treasury codes, closing tracking gaps with **100% data fidelity**.
- Constructed a **3-tier defaulter classification module** in Python to track non-compliant bonded personnel and automate reporting outputs.
- Integrated backend database connectors across MariaDB, State Treasury, and National Next-Gen e-Hospital systems, resolving schema discrepancies.
- Collaborated with National Informatics Centre (NIC) teams to optimize data extraction queries and standardize Python preprocessing utilities.

KEY PROJECTS

Proactive 5G Handover Prediction (Diffusion Models) • *M.Tech Thesis — Python & High-Throughput Telemetry Pipeline*
Technologies: Python, PyTorch, NumPy, Scikit-learn, Pandas, SMOTE, Tomek Links

- Developed an end-to-end Python pipeline pairing a **GRU encoder** with a **Diffusion Model (DDPM/DDIM)** forecasting signal drops from **spatio-temporal** telemetry.
- Addressed **98%/2% class imbalance** using **SMOTE + Tomek Links** in Python, training an ensemble Random Forest policy attaining a **97.64% F1-score**.

AI vs. Human Text Classification Pipeline • *Python Deep Learning & NLP Engineering*
Technologies: Python, TensorFlow, PyTorch, BERT, XGBoost, Scikit-learn, Regex

- Engineered dual Python classification pipelines: extracted 768-dim dense **BERT embeddings** with **XGBoost**, and built a hybrid Bi-LSTM/Transformer model.
- Implemented NFKD text normalization, dropout regularization, and dynamic learning rate decay, achieving **99% Precision** and **98% Recall**.

EDUCATION

M.Tech in Computer Science & Engineering | IIIT Naya Raipur (CGPA: 7.90) | 08/2024 – 07/2026
B.Tech in Computer Science | Amity University Chhattisgarh (CGPA: 7.71) | 08/2019 – 06/2023

CERTIFICATIONS & KEY QUALIFICATIONS

- **Qualified GATE Computer Science (CS)**, 2024
- **Python for Data Science & AI** (Coursera)
- **SQL & Database Design** (Udemy)
- **Microsoft Power BI** (Udemy)
- **Celonis Process Mining Foundations** (Credly)
- **IELTS Academic:** 7.5 Band Score